

TATTOO STUDIO APP

Tech Stack — ASP.NET Core 10 + React + MySQL + SignalR

FRONTEND

React 19 + Vite

UI library + build tool. Fully decoupled SPA, talks to .NET API over HTTP and SignalR.

TypeScript 5+

Static typing end-to-end. Strict mode enforced across the entire frontend codebase.

React Router v7

Client-side routing with role-based route guards. Each role gets scoped dashboard routes.

Redux Toolkit

Client state — current user, role, tenant context, UI state, SignalR connection state.

RTK Query

Server state — all API calls, caching, background refetch, automatic cache invalidation on mutations.

React Redux

Official React bindings for Redux. Connects components to the store.

Tailwind CSS + shadcn/ui

Utility-first styling + accessible, unstyled component primitives. Themeable per studio.

@microsoft/signalr

Official SignalR JS client. Real-time schedule updates, booking events, notifications.

React Hook Form

Form state and client-side validation. Pairs with FluentValidation on the backend.

BACKEND

ASP.NET Core 10 + C#

Web framework. Minimal API for explicit, performant endpoint definitions with full control.

MediatR

CQRS pattern — commands and queries separated cleanly. Decouples handlers from endpoints.

FluentValidation

Input validation via MediatR pipeline behaviors. Single validation layer for the entire app.

ASP.NET Core Identity

User management, password hashing, lockout policies, role management.

JWT Bearer Auth

Access tokens (15 min expiry) + refresh tokens in httpOnly cookies. Tenant ID + role in claims.

Policy-based RBAC

Four roles: client, artist, owner, issuer. Enforced at endpoint level via authorization policies.

SignalR

Built-in real-time layer. Live schedule, booking events, design approvals, notification feed.

Serilog

Structured JSON logging. Every log includes tenant_id, user_id, request_id. No free-form strings.

OpenTelemetry (.NET)

Traces and metrics. Native .NET 10 support. Feeds into Grafana Tempo and Prometheus.

DATA

MySQL 8.4 LTS

Primary relational database. LTS release ensures long-term support and stability.

EF Core 10 + Pomelo

ORM with Pomelo MySQL provider. Global query filters enforce tenant isolation on every query.

Redis (StackExchange)

Session store, rate limiting per tenant, appointment slot locking, hot read caching.

SERVICES

Stripe.net

Official C# SDK. Stripe Connect for multi-tenant payouts. Deposit collection and management.

MailKit / Resend

Transactional email. Booking confirmations, aftercare instructions, design approval links.

Twilio

SMS reminders at 48h and 24h. Aftercare messages and touch-up reminders via C# SDK.

Hangfire + Hangfire.MySql

Background job scheduling. Reminder sequences, deposit forfeiture, recurring touch-up checks.

Cloudflare R2 (AWSSDK.S3)

File storage. Consent forms, design drafts, before/after photos. Signed URLs per tenant.

INFRASTRUCTURE

Docker

Containerize .NET API and React build. Consistent environments across dev, staging, production.

K3s / Kubernetes

Lightweight Kubernetes for production orchestration. Full control over scaling and deployments.

Nginx

Reverse proxy. Serves React static build, forwards API requests, terminates TLS.

GitHub Actions

CI/CD pipeline. Build, test, lint, security scan on every PR. Deploy on merge to main.

Cloudflare

CDN, WAF, DDoS protection. Edge caching for public booking pages.

Hetzner / AWS

Primary hosting. Hetzner for cost efficiency, AWS for managed services at scale.

OBSERVABILITY

Grafana

Unified dashboards for metrics, logs, and traces. Single pane of glass for production health.

Prometheus

Metrics collection. RED method per endpoint — Rate, Errors, Duration.

Grafana Loki

Log aggregation. Structured JSON logs from Serilog indexed by tenant_id and request_id.

Grafana Tempo

Distributed tracing. Essential for debugging cross-tenant latency and SignalR issues.

QUICK REFERENCE

Frontend	React 19 · Vite · TypeScript · React Router v7 · Redux Toolkit · RTK Query · Tailwind · shadcn/ui · SignalR JS · React Hook Form
Backend	ASP.NET Core 10 · C# · Minimal API · MediatR · FluentValidation · Identity · JWT · SignalR · Serilog · OpenTelemetry
Data	MySQL 8.4 · EF Core 10 (Pomelo) · Redis
Services	Stripe.net · MailKit/Resend · Twilio · Hangfire · Cloudflare R2
Infra	Docker · K3s · Nginx · GitHub Actions · Cloudflare · Hetzner/AWS
Observability	Grafana · Prometheus · Loki · Tempo

Every tool in this stack has exactly one job. No redundant dependencies.